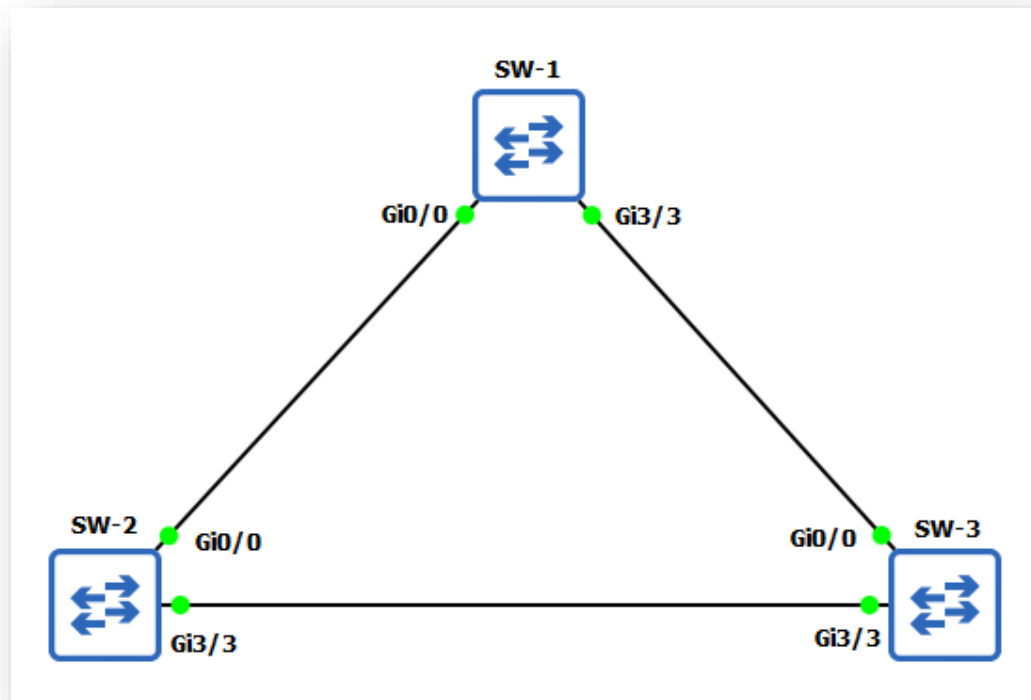


CDP (Cisco Discovery Protocol) & LLDP (Link Layer Discovery Protocol)



Task

1. Configure the SW-1 with IP 150.150.150.1, SW-2 with IP 150.150.150.2 and SW-3 with IP 150.150.150.3 for device management and hostnames.
2. Verify CDP parameters
3. Change CDP timers to 30 sec and holdtime to 90 sec on SW-1
4. Enable LLDP on all switches.
5. Change LLDP timers to 10 sec and holdtime to 40 sec on SW-1
6. On SW-1, configure LLDP such that it should get other network device information but should not send its information to another network device.

Task-1: Configure Hostname on switches as SW-1, SW-2 and SW-3. Configure the SW-1 with IP 150.150.150.1, SW-2 with IP 150.150.150.2 and SW-3 with IP 150.150.150.3 for device management.

SW-1 Configuration:

```

Switch#config t
Switch(config)#hostname SW-1
SW-1(config)#no ip domain-lookup
SW-1(config)#int vlan1
SW-1(config-if)#ip address 150.150.150.1 255.255.255.0
SW-1(config-if)#no shut
SW-1(config-if)#exit
SW-1(config)#exit
  
```

SW-1

SW-2 Configuration:

```
Switch#config t
Switch(config)#hostname SW-2
SW-2(config)#no ip domain-lookup
SW-2(config)#int vlan1
SW-2(config-if)#ip address 150.150.150.2 255.255.255.0
SW-2(config-if)#no shut
SW-2(config-if)#exit
SW-2(config)#exit
SW-2#
```

SW-3 Configuration:

```
Switch#config t
Switch(config)#hostname SW-3
SW-3(config)#no ip domain-lookup
SW-3(config)#int vlan1
SW-3(config-if)#ip address 150.150.150.3 255.255.255.0
SW-3(config-if)#no shut
SW-3(config-if)#exit
SW-3(config)#exit
SW-3#
```

Note: CDP is Cisco proprietary protocol and by default, CDP is enabled on Cisco devices. No need to enable it.

✓ Verification & Testing:**SW-1#sh cdp neighbors**

Capability Codes: R - Router, T - Trans Bridge, B - Source Route Bridge
S - Switch, H - Host, I - IGMP, r - Repeater, P - Phone,
D - Remote, C - CVTA, M - Two-port Mac Relay

Device ID	Local Intrfce	Holdtme	Capability	Platform	Port ID
SW-3	Gig 3/3	151	R S I		Gig 0/0
SW-2	Gig 0/0	125	R S I		Gig 0/0

Total cdp entries displayed : 2

SW-2#sh cdp neighbors

Capability Codes: R - Router, T - Trans Bridge, B - Source Route Bridge
S - Switch, H - Host, I - IGMP, r - Repeater, P - Phone,
D - Remote, C - CVTA, M - Two-port Mac Relay

Device ID	Local Intrfce	Holdtme	Capability	Platform	Port ID
SW-1	Gig 0/0	178	R S I		Gig 0/0
SW-3	Gig 3/3	130	R S I		Gig 3/3

Total cdp entries displayed : 2

SW-3#sh cdp neighbors

Capability Codes: R - Router, T - Trans Bridge, B - Source Route Bridge
S - Switch, H - Host, I - IGMP, r - Repeater, P - Phone,
D - Remote, C - CVTA, M - Two-port Mac Relay

Device ID	Local Intrfce	Holdtme	Capability	Platform	Port ID
SW-1	Gig 0/0	153	R S I	Gig 3/3	
SW-2	Gig 3/3	142	R S I	Gig 3/3	

Total cdp entries displayed : 2

SW-1#sh cdp neighbors detail

Device ID: SW-3

Entry address(es):

IP address: 150.150.150.3

Platform: Cisco , Capabilities: Router Switch IGMP

Interface: GigabitEthernet3/3, Port ID (outgoing port): GigabitEthernet0/0

Holdtime : 127 sec

Version :

Cisco IOS Software, vios_l2 Software (vios_l2-ADVENTERPRISEK9-M), Experimental Version
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advertisement version: 2

VTP Management Domain: "

Native VLAN: 1

Duplex: full

Management address(es):

IP address: 150.150.150.3

Device ID: SW-2

Entry address(es):

IP address: 150.150.150.2

Platform: Cisco , Capabilities: Router Switch IGMP

Interface: GigabitEthernet0/0, Port ID (outgoing port): GigabitEthernet0/0

Holdtime : 152 sec

Version :

Cisco IOS Software, vios_l2 Software (vios_l2-ADVENTERPRISEK9-M), Experimental Version
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advertisement version: 2

VTP Management Domain: "

Native VLAN: 1

Duplex: full

Management address(es):

IP address: 150.150.150.2

Total cdp entries displayed : 2

SW-2#sh cdp neighbors detail

Device ID: SW-1

Entry address(es):

IP address: 150.150.150.1

Platform: Cisco , Capabilities: Router Switch IGMP

Interface: GigabitEthernet0/0, Port ID (outgoing port): GigabitEthernet0/0

Holdtime : 163 sec

Version :

Cisco IOS Software, vios_l2 Software (vios_l2-ADVENTERPRISEK9-M), Experimental Version
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advertisement version: 2

VTP Management Domain: "

Native VLAN: 1

Duplex: full

Management address(es):

IP address: 150.150.150.1

Device ID: SW-3

Entry address(es):

IP address: 150.150.150.3

Platform: Cisco , Capabilities: Router Switch IGMP

Interface: GigabitEthernet3/3, Port ID (outgoing port): GigabitEthernet3/3

Holdtime : 171 sec

Version :

Cisco IOS Software, vios_l2 Software (vios_l2-ADVENTERPRISEK9-M), Experimental Version
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advertisement version: 2

VTP Management Domain: "

Native VLAN: 1

Duplex: full

Management address(es):

IP address: 150.150.150.3

Total cdp entries displayed : 2

SW-3#sh cdp neighbors detail**Device ID: SW-1**

Entry address(es):

IP address: 150.150.150.1

Platform: Cisco , Capabilities: Router Switch IGMP

Interface: GigabitEthernet0/0, Port ID (outgoing port): GigabitEthernet3/3

Holdtime : 151 sec

Version :

Cisco IOS Software, vios_l2 Software (vios_l2-ADVENTERPRISEK9-M), Experimental Version 15.2(20170321:233949) [mmen 101]

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advertisement version: 2

VTP Management Domain: "

Native VLAN: 1

Duplex: full

Management address(es):

IP address: 150.150.150.1**Device ID: SW-2**

Entry address(es):

IP address: 150.150.150.2

Platform: Cisco , Capabilities: Router Switch IGMP

Interface: GigabitEthernet3/3, Port ID (outgoing port): GigabitEthernet3/3

Holdtime : 153 sec

Version :

Cisco IOS Software, vios_l2 Software (vios_l2-ADVENTERPRISEK9-M), Experimental Version 15.2(20170321:233949) [mmen 101]

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advertisement version: 2

VTP Management Domain: "

Native VLAN: 1

Duplex: full

Management address(es):

IP address: 150.150.150.2**Total cdp entries displayed : 2****SW-1#sh cdp**

Global CDP information:

Sending CDP packets every 60 seconds**Sending a holdtime value of 180 seconds**

Sending CDPv2 advertisements is enabled

SW-2#sh cdp

Global CDP information:

Sending CDP packets every 60 seconds

Sending a holdtime value of 180 seconds

Sending CDPv2 advertisements is enabled

SW-3#sh cdp

Global CDP information:

Sending CDP packets every 60 seconds

Sending a holdtime value of 180 seconds

Sending CDPv2 advertisements is enabled

SW-1#sh cdp interface | section up|interfaces

GigabitEthernet0/0 is up, line protocol is up

Encapsulation ARPA

Sending CDP packets every 60 seconds

Holdtime is 180 seconds

GigabitEthernet3/3 is up, line protocol is up

Encapsulation ARPA

Sending CDP packets every 60 seconds

Holdtime is 180 seconds

cdp enabled interfaces : 16

interfaces up : 2

interfaces down : 14

SW-2#sh cdp interface | section up|interfaces

GigabitEthernet0/0 is up, line protocol is up

Encapsulation ARPA

Sending CDP packets every 60 seconds

Holdtime is 180 seconds

GigabitEthernet3/3 is up, line protocol is up

Encapsulation ARPA

Sending CDP packets every 60 seconds

Holdtime is 180 seconds

cdp enabled interfaces : 16

interfaces up : 2

interfaces down : 14

SW-3#sh cdp interface | section up|interfaces

GigabitEthernet0/0 is up, line protocol is up

Encapsulation ARPA

Sending CDP packets every 60 seconds

Holdtime is 180 seconds

GigabitEthernet3/3 is up, line protocol is up

Encapsulation ARPA

Sending CDP packets every 60 seconds

Holdtime is 180 seconds

cdp enabled interfaces : 16

interfaces up : 2

interfaces down : 14

Task-3: Change CDP timers to 30 sec and holdtime to 90 sec on SW-1**SW-1 Configuration:**

```
SW-1#config t
SW-1(config)#cdp timer 30
SW-1(config)#cdp holdtime 90
SW-1(config)#exit
SW-1#
```

✓ Verification & Testing:**SW-1#sh cdp**

Global CDP information:

Sending CDP packets every 30 seconds
Sending a holdtime value of 90 seconds
Sending CDPv2 advertisements is enabled

Disable CDP globally or on interface basis.

```
SW-1#config t
SW-1(config)#no cdp run
SW-1(config)#exit
SW-1#
```

Or

```
SW-1#config t
SW-1(config)#interface range gi0/0
SW-1(config-if)#no cdp enable
SW-1(config-if)#exit
SW-1(config)#exit
SW-1#
```

LLDP (Link Layer Discovery Protocol)**Task-4: Enable LLDP on all switches.**

```
SW-1#sh lldp
% LLDP is not enabled
SW-1#
```

Note: The Link Layer Discovery Protocol (LLDP) is a vendor-neutral link layer protocol and we need to manually enable it on network device.

SW-1 Configuration:

```
SW-1#config t
SW-1(config)#lldp run
```

SW-1(config)#exit
SW-1#

SW-2 Configuration:

SW-2#config t
SW-2(config)#lldp run
SW-2(config)#exit
SW-2#

SW-3 Configuration:

SW-3#config t
SW-3(config)#lldp run
SW-3(config)#exit
SW-3#

✓ Verification & Testing:

SW-1#sh lldp

Global LLDP Information:
Status: ACTIVE
LLDP advertisements are sent every 30 seconds
LLDP hold time advertised is 120 seconds
LLDP interface reinitialisation delay is 2 seconds

SW-2#sh lldp

Global LLDP Information:
Status: ACTIVE
LLDP advertisements are sent every 30 seconds
LLDP hold time advertised is 120 seconds
LLDP interface reinitialisation delay is 2 seconds

SW-3#sh lldp

Global LLDP Information:
Status: ACTIVE
LLDP advertisements are sent every 30 seconds
LLDP hold time advertised is 120 seconds
LLDP interface reinitialisation delay is 2 seconds

SW-1#sh lldp neighbors

Capability codes:
(R) Router, (B) Bridge, (T) Telephone, (C) DOCSIS Cable Device
(W) WLAN Access Point, (P) Repeater, (S) Station, (O) Other

Device ID	Local Intf	Hold-time	Capability	Port ID
SW-2	Gi0/0	120	R	Gi0/0
SW-3	Gi3/3	120	R	Gi0/0

Total entries displayed: 2

SW-2#sh lldp neighbors

Capability codes:

(R) Router, (B) Bridge, (T) Telephone, (C) DOCSIS Cable Device

(W) WLAN Access Point, (P) Repeater, (S) Station, (O) Other

Device ID	Local Intf	Hold-time	Capability	Port ID
SW-1	Gi0/0	120	R	Gi0/0
SW-3	Gi3/3	120	R	Gi3/3

Total entries displayed: 2

SW-3#sh lldp neighbors

Capability codes:

(R) Router, (B) Bridge, (T) Telephone, (C) DOCSIS Cable Device

(W) WLAN Access Point, (P) Repeater, (S) Station, (O) Other

Device ID	Local Intf	Hold-time	Capability	Port ID
SW-2	Gi3/3	120	R	Gi3/3
SW-1	Gi0/0	120	R	Gi3/3

Total entries displayed: 2

SW-1#sh lldp neighbors detail-----
Local Intf: Gi0/0

Chassis id: 0c17.92f5.9600

Port id: Gi0/0

Port Description: GigabitEthernet0/0

System Name: SW-2

System Description:

Cisco IOS Software, vios_12 Software (vios_12-ADVENTERPRISEK9-M), Experimental Version
15.2(20170321:233949) [mmen 101]

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Time remaining: 96 seconds

System Capabilities: B,R

Enabled Capabilities: R

Management Addresses:

IP: 150.150.150.2

Auto Negotiation - not supported

Physical media capabilities - not advertised

Media Attachment Unit type - not advertised

Vlan ID: - not advertised

Local Intf: Gi3/3

Chassis id: 0c17.929f.7d00

Port id: Gi0/0

Port Description: GigabitEthernet0/0

System Name: SW-3

System Description:

Cisco IOS Software, vios_12 Software (vios_12-ADVENTERPRISEK9-M), Experimental Version 15.2(20170321:233949) [mmen 101]

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Time remaining: 107 seconds

System Capabilities: B,R

Enabled Capabilities: R

Management Addresses:

IP: 150.150.150.3

Auto Negotiation - not supported

Physical media capabilities - not advertised

Media Attachment Unit type - not advertised

Vlan ID: - not advertised

Total entries displayed: 2

SW-2#sh lldp neighbors detail

Local Intf: Gi0/0

Chassis id: 0c17.9259.7100

Port id: Gi0/0

Port Description: GigabitEthernet0/0

System Name: SW-1

System Description:

Cisco IOS Software, vios_12 Software (vios_12-ADVENTERPRISEK9-M), Experimental Version 15.2(20170321:233949) [mmen 101]

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Time remaining: 108 seconds

System Capabilities: B,R

Enabled Capabilities: R

Management Addresses:

IP: 150.150.150.1

Auto Negotiation - not supported

Physical media capabilities - not advertised

Media Attachment Unit type - not advertised

Vlan ID: - not advertised

Local Intf: Gi3/3

Chassis id: 0c17.929f.7d00

Port id: Gi3/3

Port Description: GigabitEthernet3/3

System Name: SW-3

System Description:

Cisco IOS Software, vios_12 Software (vios_12-ADVENTERPRISEK9-M), Experimental Version 15.2(20170321:233949) [mmen 101]

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Time remaining: 100 seconds

System Capabilities: B,R

Enabled Capabilities: R

Management Addresses:

IP: 150.150.150.3

Auto Negotiation - not supported

Physical media capabilities - not advertised

Media Attachment Unit type - not advertised

Vlan ID: - not advertised

Total entries displayed: 2

SW-3#sh lldp neighbors detail

Local Intf: Gi3/3

Chassis id: 0c17.92f5.9600

Port id: Gi3/3

Port Description: GigabitEthernet3/3

System Name: SW-2

System Description:

Cisco IOS Software, vios_l2 Software (vios_l2-ADVENTERPRISEK9-M), Experimental Version
15.2(20170321:233949) [mmen 101]

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Time remaining: 110 seconds

System Capabilities: B,R

Enabled Capabilities: R

Management Addresses:

IP: 150.150.150.2

Auto Negotiation - not supported

Physical media capabilities - not advertised

Media Attachment Unit type - not advertised

Vlan ID: - not advertised

Local Intf: Gi0/0

Chassis id: 0c17.9259.7100

Port id: Gi3/3

Port Description: GigabitEthernet3/3

System Name: SW-1

System Description:

Cisco IOS Software, vios_l2 Software (vios_l2-ADVENTERPRISEK9-M), Experimental Version
15.2(20170321:233949) [mmen 101]

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Time remaining: 99 seconds

System Capabilities: B,R

Enabled Capabilities: R

Management Addresses:

IP: 150.150.150.1

Auto Negotiation - not supported

Physical media capabilities - not advertised

Media Attachment Unit type - not advertised

Vlan ID: - not advertised

Total entries displayed: 2

SW-1# sh lldp interface gigabitEthernet 0/0

GigabitEthernet0/0:

Tx: enabled

Rx: enabled

Tx state: IDLE

Rx state: WAIT FOR FRAME

SW-1# sh lldp interface gigabitEthernet 3/3

GigabitEthernet3/3:

Tx: enabled

Rx: enabled

Tx state: IDLE

Rx state: WAIT FOR FRAME

SW-2#sh lldp interface gigabitEthernet 0/0

GigabitEthernet0/0:

Tx: enabled

Rx: enabled

Tx state: IDLE

Rx state: WAIT FOR FRAME

SW-2#sh lldp interface gigabitEthernet 3/3

GigabitEthernet3/3:

Tx: enabled

Rx: enabled

Tx state: IDLE

Rx state: WAIT FOR FRAME

SW-3#sh lldp interface gigabitEthernet 0/0

GigabitEthernet0/0:

Tx: enabled

Rx: enabled

Tx state: IDLE

Rx state: WAIT FOR FRAME

SW-3#sh lldp interface gigabitEthernet 3/3

GigabitEthernet3/3:

Tx: enabled

Rx: enabled

Tx state: IDLE

Rx state: WAIT FOR FRAME

Task-5: Change LLDP timers to 10 sec and holdtime to 40 sec on SW-1

SW-1 Configuration:

```
SW-1#config t
SW-1(config)#lldp timer 10
SW-1(config)#lldp holdtime 40
SW-1(config)#exit
SW-1#
```

✓ Verification & Testing:

SW-1#sh lldp

Global LLDP Information:

Status: ACTIVE

LLDP advertisements are sent every 10 seconds

LLDP hold time advertised is 40 seconds

LLDP interface reinitialisation delay is 2 seconds

Task-6: On SW-1, configure LLDP such that it should get other network device information but should not send its information to another network device.

SW-1 Configuration:

```
SW-1#config t
SW-1(config)#interface range gi0/0, gi3/3
SW-1(config-if-range)#no lldp transmit
SW-1(config-if-range)#exit
SW-1(config)#exit
SW-1#
```

Verification & Testing:

SW-1# sh lldp interface gigabitEthernet 0/0

GigabitEthernet0/0:

Tx: disabled

Rx: enabled

Tx state: INIT

Rx state: WAIT FOR FRAME

SW-1# sh lldp interface gigabitEthernet 3/3

GigabitEthernet3/3:

Tx: disabled

Rx: enabled

Tx state: INIT

Rx state: WAIT FOR FRAME

SW-1#sh lldp neighbors

Capability codes:

(R) Router, (B) Bridge, (T) Telephone, (C) DOCSIS Cable Device
(W) WLAN Access Point, (P) Repeater, (S) Station, (O) Other

Device ID	Local Intf	Hold-time	Capability	Port ID
SW-2	Gi0/0	120	R	Gi0/0
SW-3	Gi3/3	120	R	Gi0/0

Total entries displayed: 2

SW-2#sh lldp neighbors

Capability codes:

(R) Router, (B) Bridge, (T) Telephone, (C) DOCSIS Cable Device
(W) WLAN Access Point, (P) Repeater, (S) Station, (O) Other

Device ID	Local Intf	Hold-time	Capability	Port ID
SW-3	Gi3/3	120	R	Gi3/3

Total entries displayed: 1

SW-3#sh lldp neighbors

Capability codes:

(R) Router, (B) Bridge, (T) Telephone, (C) DOCSIS Cable Device
(W) WLAN Access Point, (P) Repeater, (S) Station, (O) Other

Device ID	Local Intf	Hold-time	Capability	Port ID
SW-2	Gi3/3	120	R	Gi3/3

Total entries displayed: 1

Disable LLDP globally or on interface basis.

```
SW-1#config t
SW-1(config)#no lldp run
SW-1(config)#exit
SW-1#
```

Or

```
SW-1#config t
SW-1(config)#interface range gi0/0, gi3/3
SW-1(config-if-range)#no lldp transmit
SW-1(config-if-range)#no lldp receive
SW-1(config-if-range)#exit
SW-1(config)#exit
SW-1#
```

Important Commands:

```
sh cdp
sh cdp neighbors
sh cdp neighbors details
sh cdp interface | section up|interfaces
sh lldp
sh lldp neighbors
sh lldp neighbors detail
sh lldp interface gigabitEthernet x/x
```

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